BLE ESL AP Manual

V0.1

About this Document

This document supports “RT58x_SDK_v2.2.0” and later version.

Table of Contents

1. Introduction

This document is the command sets for implementing the BLE ESL AP function with BLE ESL AP module. It includes the BLE ESL network configuration and ESL commands for network management and application commands.

2. Hardware Interface Setup

The ESL AP module is connected to host control unit by UART port. The default baud rate is 115200 with 8-bit data length, no parity bit, and 1 stop bit format.

3. Command Data Format

The BLE ESL AP command is constructed as the following format. It uses the little endian format

Header	Length	Command ID / Event code	Parameter	Checksum
4 octets	1 octet	1 octets	n octets	1 octet

3.1 Header field

The command header is 4 bytes long and should be formatted as 0xFF 0xFC 0xFC 0xFF or 0xFF 0xFD 0xFD 0xFF.

Header	Description
0xFF 0xFC 0xFC 0xFF	BLE control and application control command
0xFF 0xFD 0xFD 0xFF	Event and response

3.2 Length field

The command data length value is the length sum of command id and parameter.

3.3 Command ID / Event code field

The command id / event code is 1 byte long, which is either the command ID or the event code depending on command header

3.4 Parameter field

The parameter is variable bytes and used for command to configure the devices or the response of devices. The following command/event description has more detail information.

3.5 Checksum field

The checksum is 1-byte long and to confirm the received data correctly. Its value is bitwise not(~) of the sum of all command data fields but header field excluded. Checksum value = ~(length[0] + command id[0] + parameter[0] + parameter[1] + ……. + parameter[n-1]).

4 BLE control command

4.1 Scan control

- Command Id
  0x01

- Parameter

Field	Size(octets)	Notes
Interval	2	Scan interval
Window	2	Scan window
Control	1	Enable scan or disable

Scan control is the command used to enable or disable the scan for advertising.

Scan interval unit is 0.625ms, the minimum scan interval is 0x0004 (0x0004 * 0.625ms = 2.5ms) and maximum scan interval is 0x4000 (0x4000 * 0.625ms 10.24s).

The unit of scan window is 0.625ms, the minimum scan window is 0x0004 (0x0004 * 0.625ms = 2.5ms) and maximum scan window is 0x4000 (0x4000 * 0.625ms = 10.24s).

Control filed 1 is to start scan the device, 0 is to stop scan the device

- Event(s):

  When scan control command has been sent, a command complete event should be received, if the status is successful, the scan advertising reports will be reported via advertising report event.

4.2 Create connection

- Command Id

  0x02

- Parameter

Field	Size(octets)	Notes
Scan interval	2	Scan interval
Scan window	2	Scan window
Connection interval	2	Connection window
Connection timeout	2	Connection timeout
Address type	1	Peer address type
Address	6	Peer address

Create connection command is used to establish a connection with specific device.

Connection interval unit is 1.25ms, the minimum connection interval is 0x0006 (0x0006 * 1.25ms = 7.5ms) and maximum connection interval is 0x0C80 (0x0C80 * 1.25ms = 4s).

The unit of connection timeout is 10ms, the minimum connection timeout is 0x000A (0x000A * 10ms = 100ms) and maximum connection timeout is 0x0C80 (0x0C80 * 10ms = 32s).

- Event(s):

  When create connection command has been sent, a command complete event should be received, if the status is successful the connection complete event will be reported when connection established.

4.3 Enhance create connection

- Command Id

0x03

- Parameter

Field	Size(octets)	Notes
Scan interval	2	Scan interval
Scan window	2	Scan window
Connection interval	2	Connection window
Connection timeout	2	Connection timeout
Address type	1	Peer address type
Address	6	Peer address
Group ID	1	The group of peer address

Enhance create connection command is used to establish a connection with specific device over PAWR.

The group ID is set to the group address of the peer address.

- Event(s):

When enhance create connection command has been sent, a command complete event should be received, if the status is successful the connection complete event will be reported when connection established.

4.4 Connection cancel

- Command Id

0x04

- Parameter

There are no parameters for this message.

Connection cancel command is used to cancel the creating connection.

- Event(s):

When connection cancel command has been sent, a command complete event should be received.

4.5 Disconnect

- Command Id
  0x05

- Parameter
  There are no parameters for this message.

  Disconnect command is the command used to disconnect current connection.

- Event(s):
  When disconnect command has been sent, a command complete event should be received. If the status is successful, the disconnect complete event will be reported when connection disconnected.

4.6 Sync configuration

- Command Id
  0x07

- Parameter

Field	Size(octets)	Notes
Periodic interval	2	The interval of periodic advertising report
Number of subevent	1	The total subevent number
Subevent interval	1	The subevent interval
Response slot delay	1	The delay before response slot
Slot spacing	1	The spacing of each response slot
Number of response slot	1	The total number of response slot

Sync configuration command is the command for configure the parameters of PAWR.

Periodic advertising interval unit is 1.25ms, the minimum periodic advertising interval is 0x0006 (0x0006 * 1.25ms = 7.5ms) and maximum periodic advertising interval is 0xFFFF (0xFFFF * 1.25ms = 81.91875s).

Number of subevents range is from 0x00 to 0x80

Subevent interval is the time between subevent and the unit is 1.25ms.

Response slot delay is the time between periodic advertising packet of the subevent and first response slot.

Slot spacing is the time between each response slots, the unit is 0.125ms and the range is 0x02 (0x02 * 0.125ms = 0.25ms) to 0xFF (0xFF * 0.125ms = 31.875ms).

The range of number of response slot is 1 to 0xFF.

- Event(s):
  When sync configuration command has been sent, a command complete event should be received.

4.7 Sync control

- Command Id
  0x08

- Parameter

Field	Size(octets)	Notes
control	1	Enable or disable the PAWR

The Sync control command is the command to enable or disable PAWR.

- Event(s):
  When sync control command has been sent, a command complete event should be received.

4.8 Set periodic advertising information transfer

- Command Id
  0x09

- Parameter

Field	Size(octets)	Notes
skip	2	The number of periodic ADV can be skip
Sync timeout	2	The maximum time not receive periodic ADV
Service data	2	The value used for application

Set periodic advertising information transfer is the command that allow peer device to obtain synchronize information about PAWR.

Skip define as the number of periodic advertising packets may be missed since last successful receive, the range is from 0 to 0x01F3.

Sync timeout is means the synchronization timeout for PAWR, the unit is 10ms, and the range is 0x000A (0x000A * 10ms = 100ms) to 0x4000 (0x4000 * 10ms = 163.84s).

Service data provided by application.

- Event(s):

When set periodic advertising information transfer command has been sent, a command complete event should be received.

5 Application control command

5.1 ESL AP configuration

- Command id

0x20

- Parameter

Field	Size(octets)	Notes
AP sync key	16	AP sync key value
AP sync nonce	8	AP sync nonce value
ESL response key	16	ESL Response key value
ESL response nonce	8	ESL Response nonce value

The AP sync key and AP sync nonce used by the ESL AP as an input to the encrypted message which send to ESL devices.

The ESL response key and ESL response nonce are the input information used by the ESL devices to send the encrypted message to ESL AP.

- Event(s):

When ESL AP configuration command has been sent, a command complete event should be received.

5.2 ESL tag configuration

- Command id

0x21

- Parameter

Field	Size(octets)	Notes
Group ID	1	The Group of ESL
ESL ID	1	The unique ID within each group

ESL address consist of a 7-bit group ID and an 8-bit ESL ID.

- Event(s):

When ESL tag configuration command has been sent, a command complete event should be received.

5.3 ESL image information

- Command id

0x22

- Parameter

Field	Size(octets)	Notes
Image index	1	The index of image
Part	1	The part of image
Image length	2	Image length

The image index is the unique index for the image.

The Part field is used for same image index, an image can be combined with several parts such as the black part or read part.

The image length for following image.

- Event(s):

When ESL image information command has been sent, a command complete event should be received. If the status is successful, the ESL image information response will be reported.

5.4 ESL image transfer

- Command id

0x23

- Parameter

Field	Size(octets)	Notes
Image data	n	Image data

Maximum image data length is 240. The first two bytes of the first image data need to contain the image length.

- Event(s):

When ESL image transfer command has been sent, a command complete event should be received. If the status is successful, the ESL image transfer response will be reported.

5.5 ESL ping

- Command id

0x24

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list

ESL ID[n]	1*n	ESL ID list

Over encrypted ADV filed is used to distinguish between the ping command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of ESL ping command is fixed as 0x00.

ESL ID field is the ESL ID of target ESL device.

- Event(s):

When ESL ping command has been sent, a command complete event should be received. If the status is successful, the basic response will be reported.

5.6 Unassociated from AP

- Command id

0x25

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list

Over encrypted ADV filed is used to distinguish between the unassociated from AP command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of unassociated from AP command is fixed as 0x01.

ESL ID field is the ESL ID of target ESL device.

- Event(s):

When unassociated from AP command has been sent, a command complete event should be received. If the status is successful, the basic response will be reported.

5.7 Service reset

- Command id

0x26

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list

Over encrypted ADV filed is used to distinguish between the service reset command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of service reset command is fixed as 0x02.

ESL ID field is the ESL ID of target ESL device.

- Event(s):

When service reset command has been sent, a command complete event should be received. If the status is successful, the basic response will be reported.

5.8 Factory reset

- Command id

0x27

- Parameter

Field	Size(octets)	Notes

Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list

Over encrypted ADV filed is used to distinguish between the factory reset command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of factory reset command is fixed as 0x03.

ESL ID field is the ESL ID of target ESL device.

- Event(s):

When factory reset command has been sent, a command complete event should be received.

5.9 Update complete

- Command id

0x28

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list

Over encrypted ADV filed is used to distinguish between the update complete command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of update complete command is fixed as 0x04.

ESL ID field is the ESL ID of target ESL device.

- Event(s):
  When factory reset command has been sent, a command complete event should be received.

5.10 Read sensor data

- Command id
  0x29

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list
Sensor index[n]	1*n	Sensor index list

Over encrypted ADV filed is used to distinguish between the read sensor data command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of read sensor data command is fixed as 0x10.

ESL ID field is the ESL ID of target ESL device.

Sensor index is the index of sensor on ESL device.

- Event(s):
  When read sensor data command has been sent, a command complete event should be received. If the status is successful, the sensor data response will be reported.

5.11 Refresh display

- Command id
  0x2A

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list
Display index[n]	1*n	Display index list

Over encrypted ADV filed is used to distinguish between the refresh display command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of refresh display command is fixed as 0x11.

ESL ID field is the ESL ID of target ESL device.

Display index is the index of display on ESL device.

- Event(s):
  When refresh display command has been sent, a command complete event should be received. If the status is successful, the display state response will be reported.

5.12 Display image

- Command id
  0x2B

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list

ESL ID[n]	1*n	ESL ID list
Display index[n]	1*n	Display index list
Image index[n]	1*n	Image index list

Over encrypted ADV filed is used to distinguish between the display image command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of display image command is fixed as 0x20.

ESL ID field is the ESL ID of target ESL device.

Display index is the index of display on ESL device.

Image index is the index of image.

- Event(s):

When display image command has been sent, a command complete event should be received. If the status is successful, the display state response will be reported.

5.13 Display timed image

- Command id

0x2C

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list
Display index[n]	1*n	Display index list
Image index[n]	1*n	Image index list
Absolute time[n]	4*n	Absolute time list

Over encrypted ADV filed is used to distinguish between the display timed image command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of display timed image command is fixed as 0x60.

ESL ID field is the ESL ID of target ESL device.

Display index is the index of display on ESL device.

Image index is the index of image.

Absolute time filed is the time when the display state change.

- Event(s):

  When display timed image command has been sent, a command complete event should be received. If the status is successful, the display state response will be reported.

5.14 LED control

- Command id

  0x2D

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list
LED index[n]	1*n	LED index list
Color[n]	1*n	Color list
Pattern[n]	7*n	Pattern list
Repeat[n]	2*n	Repeat list

Over encrypted ADV filed is used to distinguish between the display timed image command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of display timed image command is fixed as 0xB0.

ESL ID field is the ESL ID of target ESL device.

LED index is the index of LED on ESL device.

Color filed is consist of following parameters used to control multi-color LEDs.

Field	Size(bits)	Notes
Color_Red	2	Red component on or off
Color_Green	2	Green component on or off
Color_Blue	2	Blue component on or off
Brightness	2	0: 25% 1: 50% 2: 75% 3: 100%

Pattern indicates the flashing pattern setting of the LED.

Repeat filed is consist of two parameters

Field	Size(bits)	Notes
Repeat type	1	Repeat type
Repeat duration	15	Repeat duration

The definition of repeat duration is depending on repeat type. An integer number of repeats of flashing pattern if repeat type is 0, otherwise, a time duration specified in units of seconds.

- Event(s):

When LED control command has been sent, a command complete event should be received. If the status is successful, the LED state response will be reported.

5.15 LED timed control

- Command id

  0x2E

- Parameter

Field	Size(octets)	Notes
Over encrypted ADV	1	Command over EADV or not
Group ID	1	Group ID
Opcode[n]	1*n	Opcode list
ESL ID[n]	1*n	ESL ID list
LED index[n]	1*n	LED index list
Color[n]	1*n	Color list
Pattern[n]	7*n	Pattern list
Repeat[n]	2*n	Repeat list
Absolute time[n]	4*n	Absolute time list

Over encrypted ADV filed is used to distinguish between the display timed image command sent via encrypted ADV or BLE ESL service.

Group ID field is the group of target ESL device.

Opcode filed of display timed image command is fixed as 0xF0.

ESL ID field is the ESL ID of target ESL device.

LED index is the index of LED on ESL device.

Color filed is consist of following parameters used to control multi-color LEDs.

Field	Size(bits)	Notes
Color_Red	2	Red component on or off
Color_Green	2	Green component on or off
Color_Blue	2	Blue component on or off
Brightness	2	0: 25% 1: 50% 2: 75%

		3: 100%

Pattern indicates the flashing pattern setting of the LED.

Repeat filed is consist of two parameters

Field	Size(bits)	Notes
Repeat type	1	Repeat type
Repeat duration	15	Repeat duration

The definition of repeat duration is depending on repeat type. An integer number of repeats of flashing pattern if repeat type is 0, otherwise, a time duration specified in units of seconds.

Absolute time filed is the time when the LED state change.

- Event(s):

When LED timed control command has been sent, a command complete event should be received. If the status is successful, the LED state response will be reported.

5.16 OTA image

- Command id

0x2F

- Parameter

Field	Size(octets)	Notes
Address	4	Image address
Image data	n	Image data

Address field defines the flash address where the image data is stored

Image data is the upgrading image data.

- Event(s):

When OTA image command has been sent, a command complete event should be received.

5.17 OTA control

- Command id

  0x30

- Parameter

Field	Size(octets)	Notes
OTA command	1	OTA command
command data	n	Image data

OTA command defines the following commands:

Field	Value	Note
OTA command query	0x00	Query current system information
OTA command start	0x01	Start FW OTA upgrade
OTA command erase	0x02	Erase legacy FW
OTA command apply	0x03	Apply new FW

command data is the payload of different command.
OTA command query and erase did not contain any command data and command data of others command were as below:
OTA command start:

Field	Size(octets)	Note
FW length	4	The length of new FW
FW CRC	4	The CRC of new FW
Notify interval	4	The interval to send notification
FW information	1	1 = New FW was compressed 2 = New FW need check signature 3 = New FW was compressed and also need check signature
Signature length	3	The FW length for calculate signature when the FW is compressed.

OTA command apply:

Field	Size(octets)	Note

Start address	4	The starting address for storing the new FW.

- Event(s):
  When OTA control command has been sent, a command complete event should be received. If the status is successful, the OTA control response will be reported.

6 Event & response

6.1 Command complete event

- Event code
  0x01

- Parameter

Field	Size(octets)	Notes
Error code	1	The result of the command processing defined as 7.2

6.2 Advertising report event

- Event code
  0x01

- Parameter

Field	Size(octets)	Notes
Address type	1	0 = Public device address 1 = Random device address 2 = Public identity address 3 = Random identity address
Address	6	Address of advertising device

6.3 Advertising report event

- Event code
  0x02

● Parameter

Field	Size(octets)	Notes
Address type	1	0 = Public device address 1 = Random device address 2 = Public identity address 3 = Random identity address
Address	6	Address of advertising device

6.4 Connection complete event

● Event code

0x03

● Parameter

Field	Size(octets)	Notes
Status	1	Connection status
Address type	1	0 = Public device address 1 = Random device address 2 = Public identity address 3 = Random identity address
Address	6	Address of peer device
Interval	2	Connection interval
Latency	2	Peripheral latency
Timeout	2	Connection supervision timeout

6.5 Disconnection complete event

● Event code

0x04

● Parameter

Field	Size(octets)	Notes
Status	1	Disconnection status
Reason	1	Reason for disconnect

6.6 Basic state response

- Event code
  0x05

- Parameter

Field	Size(octets)	Notes
Group ID	1	Group ID of peer device
ESL ID	1	ESL ID of peer device
Basic state	16	Bit 0 = Service needed Bit 1 = Synchronized Bit 2 = Active LED Bit 3 = Pending LED update Bit 4 = Pending Display update Bit 5 – 15 = RFU

6.7 Sensor value response

- Event code
  0x06

- Parameter

Field	Size(octets)	Notes
Group ID	1	Group ID of peer device
ESL ID	1	ESL ID of peer device
Sensor index	1	Sensor index on the peer device
Sensor value	n	Sensor value of the sensor index

6.8 Display state response

- Event code
  0x07

- Parameter

Field	Size(octets)	Notes
Group ID	1	Group ID of peer device

ESL ID	1	ESL ID of peer device
Display index	1	Display index on the peer device
Image index	1	Image index of image associated with display index parameter

6.9 LED state response

- Event code

0x08

- Parameter

Field	Size(octets)	Notes
Group ID	1	Group ID of peer device
ESL ID	1	ESL ID of peer device
LED index	1	LED index on the peer device

6.10 Error response

- Event code

0x09

- Parameter

Field	Size(octets)	Notes
Group ID	1	Group ID of peer device
ESL ID	1	ESL ID of peer device
Error code	1	An error code defined as 7.3

6.11 OTA control response

- Event code

0x0A

- Parameter

Field	Size(octets)	Notes
OTA command	1	Specifics which OTA command the response is responding to.

OTA command error code	1	Error code of OTA command
Response data	n	response data has variable payload for each command with different error code, the response data were defined as

6.12 OTA download response

- Event code

0x0B

- Parameter

Field	Size(octets)	Notes
OTA notify code	1	OTA notification code
OTA notify data	n	OTA notify data has variable payload for each OTA notify code. OTA notify data were define as below table.

OTA notify code	OTA notify code description	OTA notify data	OTA notify data size(octets)
0x00	Periodic notification	Expect address	3
0x01	FOTA data was not received in a specific interval	None	-
0x02	Received FOTA data's address was not continuously	Expect address	3
0x03	Received FOTA data length incorrect	Length of current OTA data	1
0x04	Total received FOTA data length is larger than bank size	Length of total received OTA data	4
0x05	Received FOTA data's address is larger than	None	-

	bank size		
0x06	FOTA data received but FOTA procedure was not start	None	-

6.13 ESL image information response

- Event code
  0x0C

- Parameter

Field	Size(octets)	Notes
Error code	1	Error code for ESL image information command
MTU size	1	Current MTU size

6.14 ESL image transfer complete event

- Event code
  0x0D

- Parameter

Field	Size(octets)	Notes
Complete number	1	Complete number of ESL image transfer command

7 Error code

7.1 Error code of BLE control & Application control

Value	Description
0	Success
1	Unknown type
2	Not initial
3	Duplicate init
4	Data malloc fail
5	Queue malloc fail

6	Thread malloc fail
7	Semaphore malloc fail
8	Wrong configuration
9	Busy
10	SENDTO Pointer NULL
11	SENDTO fail
12	RECVFROM Pointer NULL
13	RECVFROM no data
14	RECVFROM fail
15	RECVFROM length error
16	RECVFROM memory fail
17	TIMER OP
18	Invalid state
19	Invalid parameter
20	Command not supported
21	Invalid Host ID
22	Invalid BLE handle
23	Host peripheral database parsing is still in progress
24	Detecting a sequential protocol violation. Usually happens in there is an another GATT request already in progress please wait and retry
25	BLE tx command queue full
26	BLE tx data queue full
27	BLE resource is not enough
28	BLE l2cap data channel is not exist
29	BLE l2cap data destination id is not match

7.2 Error Code of command complete event

Value	Description
0x00	Success
0x01	Unknown HCI command
0x02	Unknown connection identifier

0x03	Hardware failure
0x04	Page timeout
0x05	Authentication failure
0x06	PIN or key missing
0x07	Memory capacity exceeded
0x08	Connection timeout
0x09	Connection limit exceeded
0x0A	Synchronous connection limit to a device exceeded
0x0B	Connection already exists
0x0C	Command disallowed
0x0D	Connection rejected due to limited resources
0x0E	Connection rejected due to security reasons
0x0F	Connection rejected due to unacceptable BD_ADDR
0x10	Connection accept timeout exceeded
0x11	Unsupported feature or parameter value
0x12	Invalid HCI command parameters
0x13	Remote user terminated connection
0x14	Remote device terminated connection due to low resources
0x15	Remote device terminated connection due to power off
0x16	Connection terminated by local host
0x17	Repeated attempts
0x18	Pairing not allowed
0x19	Unknown LMP PDU
0x1A	Unsupported remote feature
0x1B	SCO offset rejected
0x1C	SCO interval rejected
0x1D	SCO air mode rejected
0x1E	Invalid LMP parameters / invalid LL parameters
0x1F	Unspecified error
0x20	Unsupported LMP parameter value / unsupported LL parameter value
0x21	Role change not allowed

0x22	LMP response timeout / LL response timeout
0x23	LMP error transaction collision / LL procedure collision
0x24	LMP PDU not allowed
0x25	Encryption mode not acceptable
0x26	Link Key cannot be changed
0x27	Requested QoS not supported
0x28	Instant passed
0x29	Pairing with unit key not supported
0x2A	Different transaction collision
0x2C	QoS unacceptable parameter
0x2D	QoS rejected
0x2E	Channel classification not supported
0x2F	Insufficient security
0x30	Parameter out of mandatory range
0x32	Role switch pending
0x34	Reserved slot violation
0x35	Role switch failed
0x36	Extended inquiry response too large
0x37	Secure simple pairing not supported by host
0x38	Host busy - pairing
0x39	Connection rejected due to no suitable channel found
0x3A	Controller busy
0x3B	Unacceptable connection parameters
0x3C	Advertising timeout
0x3D	Connection terminated due to MIC failure
0x3E	Connection failed to be established / synchronization timeout
0x3F	MAC connection failed
0x40	Coarse clock adjustment rejected but will try to adjust using clock dragging
0x41	Type0 submap not defined
0x42	Unknown advertising identifier
0x43	Limit reached
0x44	Operation cancelled by host

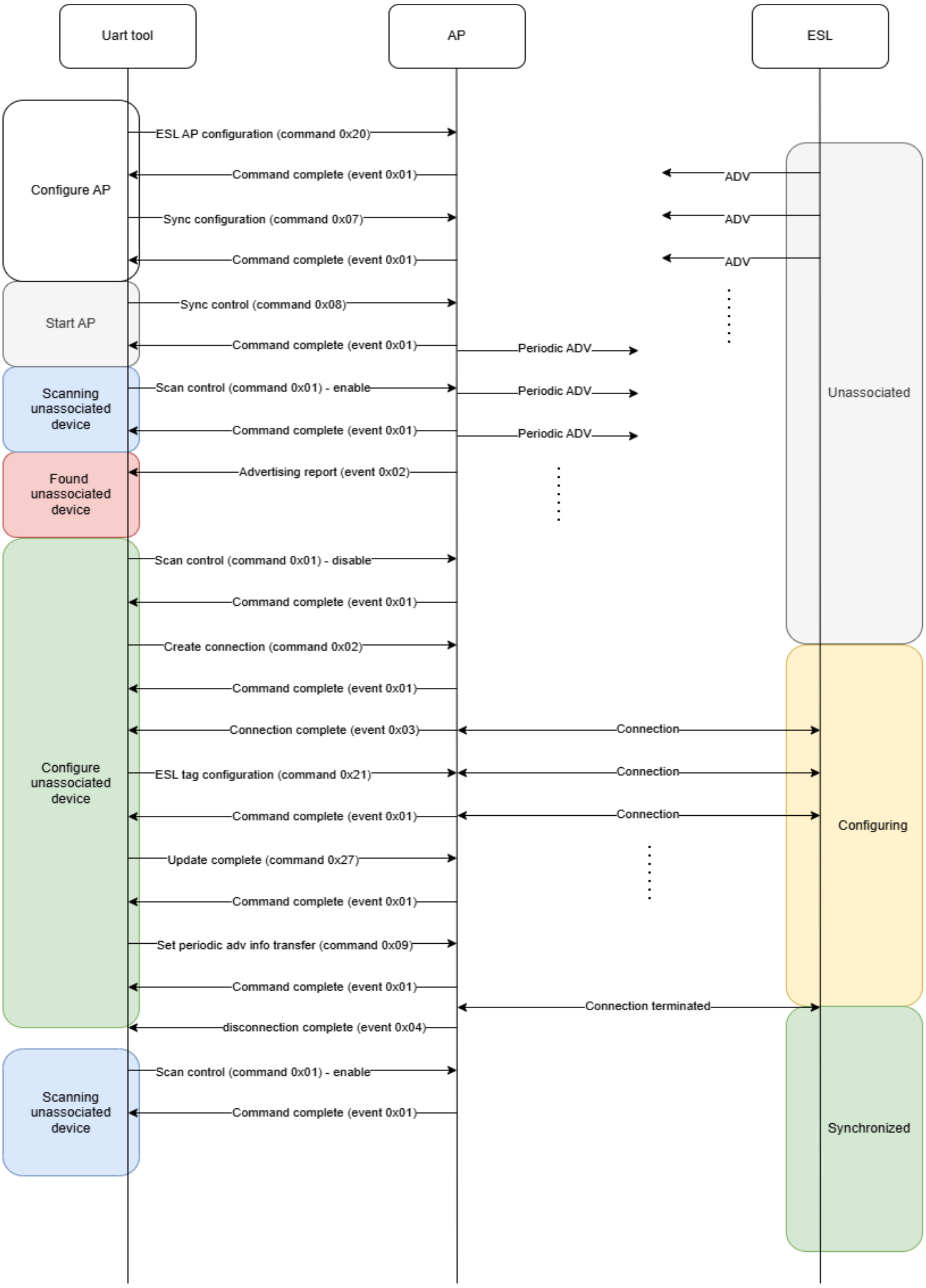
0x45	Packet too long
0xFE	Connection parameter update rejected
0xFF	Command Timeout

7.3 Error Code of error response

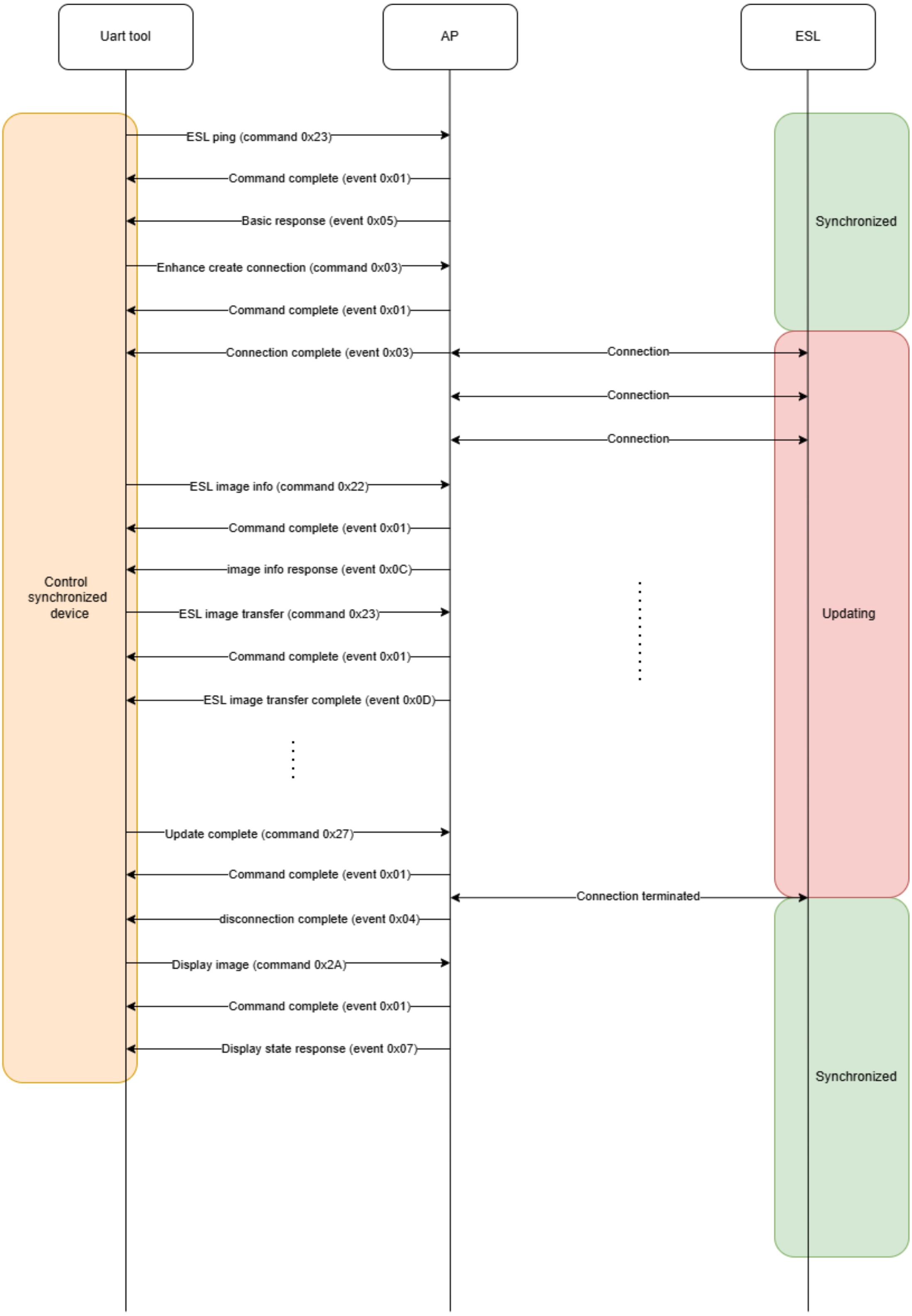
Value	Description
0x00	RFU
0x01	Unspecified error
0x02	Invalid opcode
0x03	Invalid state
0x04	Invalid image index
0x05	Image not available
0x06	Invalid parameters
0x07	Capacity limit
0x08	Insufficient battery
0x09	Insufficient resources
0x0A	Retry
0x0B	Queue full
0x0C	Implausible absolute time

8 Command flow

8.1 Join ESL device into ESL network



8.2 Control synchronized ESL device


Uart tool
AP
ESL
ESL ping (command 0x23)
Command complete (event 0x01)
Basic response (event 0x05)
Enhance create connection (command 0x03)
Command complete (event 0x01)
Connection complete (event 0x03)
Connection
Connection
Connection
ESL image info (command 0x22)
Command complete (event 0x01)
image info response (event 0x0C)
ESL image transfer (command 0x23)
Command complete (event 0x01)
ESL image transfer complete (event 0x0D)
Update complete (command 0x27)
Command complete (event 0x01)
Connection terminated
disconnection complete (event 0x04)
Display image (command 0x2A)
Command complete (event 0x01)
Display state response (event 0x07)
Control synchronized device
Synchronized
Updating
Synchronized

Revision History

Revision	Description	Owner	Date
0.1	1. Initial version.	Nat	2026/01/06